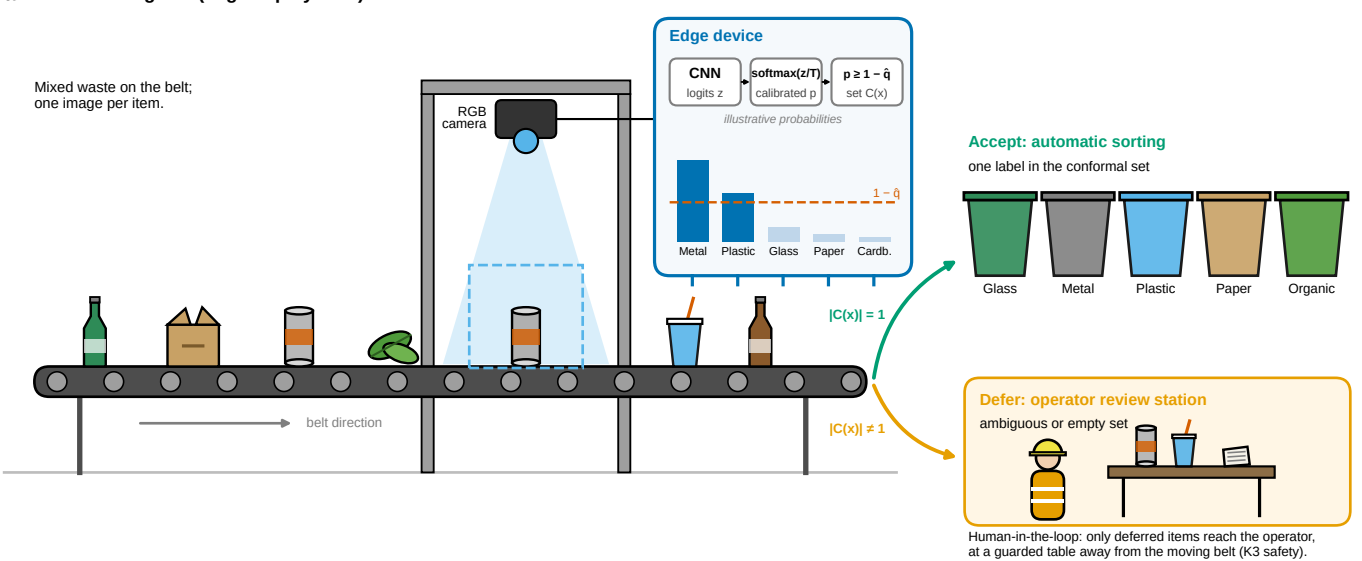

b Before deployment: leakage-aware development

1 Duplicate audit
MD5 · pHash/dHash · DINOv2
→ duplicate groups

2 Group-stratified split
train 64 · calibration-fit 8 ·
conformal-calibration 8 · test 20 %;
no group crosses roles

3 Train and calibrate
six architectures, 5 folds × 3 seeds;
T fitted on the calibration-fit
split only

4 Conformal threshold
$\hat{q}$ from scores on the
conformal-calibration split
($\alpha = 0.10$)

deploy model, T, $\hat{q}$
to the edge device (a)

Illustrative vector schematic; probabilities are illustrative, not model output.